MakeNoise<AtomDomain<T>, AbsoluteDistance<T>, MO> for ConstantTimeGeometric<T>

Michael Shoemate

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `MakeNoise` over scalars for `ConstantTimeGeometric` in `mod.rs` at commit f5bb719 (outdated[1]).

The intuition of this implementation is that a vector-valued mechanism can be used to privatize a scalar-valued input, by transforming the input into a singleton vector, applying the vector mechanism, and then unpacking the resulting singleton vector.

1 Hoare Triple

Precondition

Compiler-Verified

`MakeNoise` is parameterized as follows:

- `DI` is of type `AtomDomain<T>`
- `MI` is of type `AbsoluteDistance<T>`
- `MO` implements trait `Measure`

The following trait bounds are also required:

- Generic `T` implements trait `Integer`
- Generic `MO` implements trait `Measure`
- Type `usize` implements trait `ExactIntCast<T>`
- Type `RBig` implements trait `TryFrom<T>`
- Type `ZExpFamily<1>` implements trait `NoisePrivacyMap<L1Distance<RBig>, MO>`. This bound requires that it must be possible to construct a privacy map for the combination of `ZExpFamily<1>` noise distribution, distance type and privacy measure. Since the `ConstantTimeGeometric` distribution is equivalent to `ZExpFamily<1>`, maps built for `ZExpFamily<1>` can be used for `ConstantTimeGeometric`.

User-Verified

None

[1] See new changes with `git diff f5bb719..056354a rust/src/measurements/noise/distribution/geometric/mod.rs`

Pseudocode

```
# analogous to impl MakeNoise<AtomDomain<T>, AbsoluteDistance<T>, MO> for
    ConstantTimeGeometric<T> in Rust
class ConstantTimeGeometric:
    def make_noise(
        self, input_space: tuple[AtomDomain[T], AbsoluteDistance[QI]]
    ) -> Measurement[AtomDomain[T], T, AbsoluteDistance[QI], MO]:
        t_vec = make_vec(input_space)  #
        m_noise = self.make_noise(t_vec.output_space())  #

        return t_vec >> m_noise >> then_index_or_default(0)  #
```

Postcondition

Theorem 1.1. For every setting of the input parameters (`self, input_space, T, MO`) to `make_noise` such that the given preconditions hold, `make_noise` raises an error (at compile time or run time) or returns a valid measurement. A valid measurement has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Privacy guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an error, and `privacy_map`(`d_in`) = `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

Proof. Neither constructor `make_vec` nor `MakeNoise`.`make_noise` have manual preconditions, and the postconditions guarantee a valid transformation and valid measurement, respectively. `then_index_or_default` also does not have preconditions, and its postcondition guarantees that it returns a valid postprocessor.

The chain of a valid transformation, valid measurement and valid postprocessor is a valid measurement. □